· Users can be collected into groups on an organizational level to make them easier to administer. LDAP has built-in support for managing user groups.

· RBAC collects related sets of permissions on objects into roles which can then be assigned to users or groups and later revoked. Role assignments may be either static or dynamic.

· Roles are often specific to an API, while groups are more often defined statically for a whole organization

· ABAC evaluates access control decisions dynamically based on attributes of the subject, the resource they are accessing, the action they are attempting to perform, and the environment or context in which the request occurs (such as the time or location).

· ABAC access control decisions can be centralized using a policy engine. The XACML standard defines a common model for ABAC architecture, with separate components for policy decisions (PDP), policy information (PIP), policy administration (PAP), and policy enforcement (PEP).

ANSWERS TO EXERCISES

1. True - many group models allow groups to contain other groups as discussed in section 8.1.
2. a, c, d. Static and dynamic groups are standard, and virtual static groups are non-standard but widely implemented.
3. d - `groupOfNames` (or `groupOfUniqueNames`).